



Classification and comparison of ad hoc networks: A review

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ABSTRACT

The study of ad hoc networks and their different varieties, including wireless sensor networks, wireless mesh networks, and mobile ad hoc networks, is discussed in this paper. Wireless devices link to one another directly in an ad hoc network without the use of a central Wireless Access Point (WAP). Data transit between wireless devices is frequently managed and directed by a base station or WAP device. As soon as two or more devices converse with one another, an ad hoc network is instantaneously created. Although placing these devices close to one another is preferred, the network's quality and speed will decline as more devices join it. The idea of a vehicular ad hoc network, an intelligent vehicular ad hoc network, and a mobile ad hoc network is also introduced in ad hoc networks since nodes travel from one point to another. Any node can join and exit this network at any time. Nodes are mobile systems or gadgets that are a part of the network, like a mobile phone, laptop, personal digital assistant, MP3 player, or personal computer. Apart from this, the paper also discusses the most important results from 2000 to 2022 from various authors. In addition, the paper contrasts mobile ad hoc networks, wireless sensor networks, and wireless mesh networks in terms of their uses, limitations, issues, characteristics, and advantages and disadvantages of routing protocols. The most important things for ad hoc networks are that the different types of nodes (end-to-end nodes, intermediate nodes, and wireless antennas) are safe and respond quickly.

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Nomenclature

LNA	Local Area Network	OSPF	Open Shortest Path First routing
P2P	Peer to Peer	FSR	Fisheye State Routing
SSID	Same Service Set Identifier	TORA	Temporally Ordered Routing Algorithm
WAP	Wireless Access Point	ZRP	Zone Routing Protocol
PC	Personal Computer	RRP	Round-Robin protocol
IP	Internet Protocol	TOD	Try-once-discard
RADIUS	Remote Authentication Dial-in User Service	MEF	Maximum-Error First
AP	Access Point	CAN	Controller area network
ITS	Intelligent Transportation System	WiDom	Wireless Dominance
QoS	Quality of Service	ALOHA	Advocates of Linux open-source Hawaii association
SPANs	Smart Phone Ad Hoc Network	CSMA	Carrier sense multiple access
OLSR	Optimized Link State Routing	MAC	Media access control
FANET	Flying Ad Hoc Network	RACH	Random-access channel
IoT	Internet of Things	FDMA	Frequency division multiple access
DSRC	Dedicated Short Range Communication	TDMA	Time-division multiple access
V2V	Vehicle to Vehicle	PANs	Personal Area Networks
RSU	Road Side Unit	BANs	Body Area Networks
V2I	Vehicles to Infrastructure	VANs	Vehicle Area Network
InVANET	Intelligent Vehicular Ad Hoc Network	WLANs	Wireless Local Area Networks
LPWAN	Low Power Wide Area Network	UMTS	Universal Mobile Telecommunications System
IFC	In-Flight Wi-Fi Connecting	ODMA	Opportunity Driven Multiple Access
AANET	Aeronautical Ad Hoc Network	ABR	Associativity-Based Routing
VOIP	Voice over Internet Protocol	LAR	Location Aided Routing
AODV	Ad-Hoc On-Demand Distance Vector	WRP	Wireless Routing Protocol
DSR	Dynamic Source Routing		
DSDV	Destination Sequenced Distance Vector		

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1. Introduction

Wireless ad hoc networks (WANET) and mobile ad hoc networks (MANET) are two types of wireless networks that don't need to be set up before they can be used. It doesn't have to be fixed in one place. There aren't any routers or access points in the network in this case. This is called an ad hoc network which thinks of routing as a group effort; each node helps bypass data onto other nodes [1]. This mode is called that. The nodes can move around freely. These wireless networks don't need to be set up or run by people. They all have to send traffic that isn't related to what they're used for, so they must be used. Building a MANET is the most challenging part. Ensuring each device can always get the information to direct traffic is problematic appropriately. These networks can be separate from the more considerable internet, or they can be linked to it. Between two nodes, there may be one or more transceivers that are all different from each other. As a result, the topology changes a lot and is very self-sufficient.

Links make up the link layer of networks, and these networks are known as MANETs. Router networks are usually found on top of them [2]. There are two types of ad hoc networks. One is called an ad-hoc wireless network, and the other is a wireless point-to-point network. In this case, users can set up a type of local area network (LAN) that can be set up on the spot. In ad hoc mode, each device in a Wi-Fi network sends data to other devices. When people set up their network, there are no servers that share files and print. To connect one's devices wirelessly, users can use a basic peer-to-peer (P2P) connection to get them to talk to each other and share resources [3]. WANET has a lot of devices called "nodes" that work together to do things like routing, security, address, and key management. People in an ad hoc network need wireless adapters or chips to act as wireless routers. They must be set up independently, not as part of a network already set up.

SSID and wireless frequency channel numbers must be the same on all wireless adapters to work with each other [4]. A wireless base station doesn't make sense to ensure that messages get to each node in a network at the right time and in the correct order. In an ad hoc network, each node sends packets from and to each other instead. A wireless structure isn't built into a lot of places. For example, they can be helpful if there are no access points or routers nearby and cabling can't be used to get to where more wireless communication is needed. One is called "ad hoc," and the other is called "infrastructure," but they're not the same. Wi-Fi routers are used a lot by people who want to set up and manage Wi-Fi networks in the "infrastructure mode." Wireless access points (WAPs)

and wireless controllers are here. Ad hoc networks are made up on the spot; laptops or other devices, such as tablets or phones, are made up. Often, ad hoc networks don't use the more advanced network protocols and services found on networks set up by an organization [5]. Fig. 1 shows how this gets more difficult as the size of the MANET grows because of points.

There are times when it's better to use ad hoc mode and times when it's better to use the infrastructure. It might be better for someone who wants to set up an instant wireless network with just a few devices, but not for someone who wants to set up a long-term network with many devices [6]. The term "ad hoc" is used a lot in some new types of wireless engineering that use a lot of them. The good thing about them is that they don't need to be set up and can be set up quickly in case of emergencies like natural disasters or war. These networks can be set up soon because of this. There are many ways to make a small wireless network without building a web first. People can set up improvisation and on-the-spot networks to make a cheap all-wireless network. Infrastructure mode networks also have backup plans if some equipment breaks down. These backup plans can be used for a short time [7]. Fig. 2 shows the main vital facts about wireless ad hoc networks.

Many people use an ad-hoc wireless network to connect multiple wireless devices to the internet through an ad-hoc intermediary device that doesn't belong to anyone else. It's a computer or laptop with a wired internet connection and a wireless chip or antenna that can connect to other wireless-capable devices and share the internet with other devices. When there is no fixed network infrastructure, each network node is willing to send packets to other network nodes, even if they don't want to [8].

Distributed networks are the type of networks that this one is, and they are called such. The two or more wireless clients talk to each other without the help of a wireless access point or any other network infrastructure, like a router, in an ad-hoc wireless network. Ad hoc networks can put their safety at risk, so they should be aware of these risks. This word is used for networks that have to be set up on the spot: "ad hoc." Researchers can say this is not new because the military has been interested in ad hoc wireless networks for a long time [9]. This is happening again because of new applications and better technology.

Ad hoc networks must have a peer-to-peer architecture, and the network's topology is based on how many people move around in the same place. Cell phones have a limited range, so they might also need help from other phones to get a packet to its destination. When two mobile hosts connect, there could be a lot of wireless hops. The unpredictable network structure, decentralized control, and many-hop connections make it hard to ensure that high-speed communications are reliable in ad hoc wireless networks because there are many different ways to get to the same place [10].

Many people are considering making wireless networks set up on the fly work right now. There aren't many suitable routing protocols for wired networks because they either can't keep up with changes in the network's topology quickly or have too much overhead to keep up with them quickly. Network topologies can be rapidly changed without wasting money with dynamic source and associativity-based routing, two ways to share routing. Using rate-adaptive compression, power control, and radio clustering can help solve some of the problems using ad hoc wireless networks to run multimedia applications [11]. There are a lot of trust sources, so it's hard to come up with resource- or energy-efficient authentication procedures when there are so many of them.

Multicast and broadcast services that use a lot of trust sources simultaneously are even more likely to have this happen. There are many ways that people make their networks. They connect and talk to each other on their own. It is a Latin word that means "for this," which means something made up or made up on the

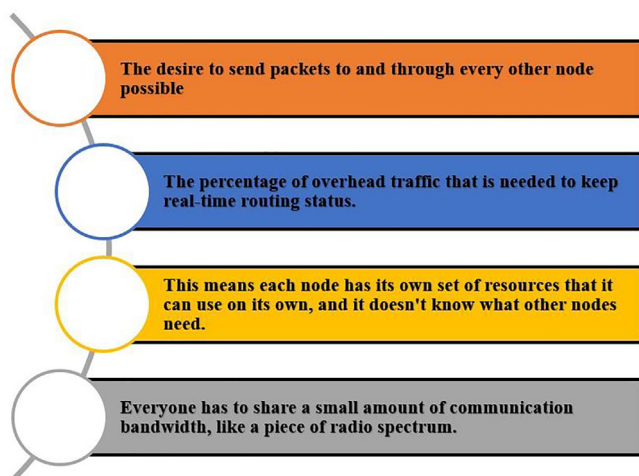


Fig. 1. Key Points to use of MANET.

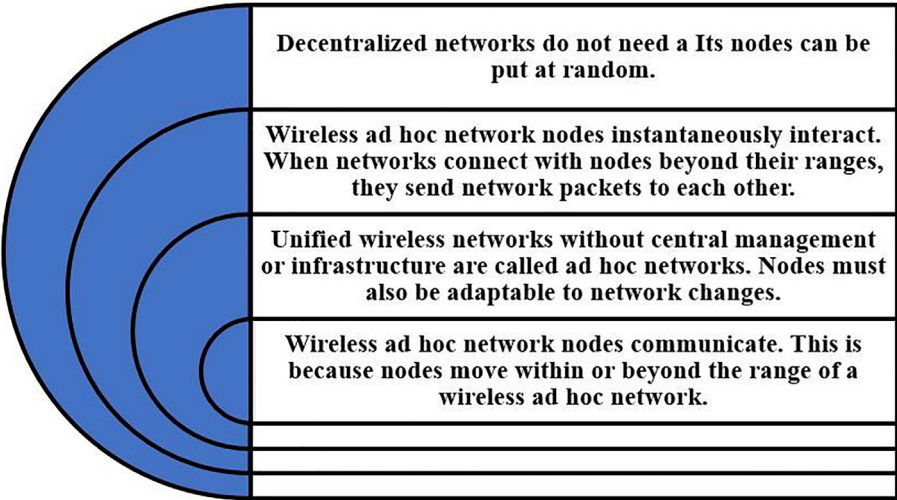


Fig. 2. Characteristics of Wireless Ad Hoc Network.

spot. Wireless LANs are the most common type of ad-hoc network in the most common type of network. There is no need to have a base station or access points when using wireless LANs. Instead, the devices talk to each other right away [12]. It is the most common way researchers connect our wireless devices. This means that all machines are connected to the same wireless Router, usually called a “Wi-Fi router.” Because they have a router, they can talk to each other on the network. When researchers send files from one device to another, there’s still a router in the middle of the process.

People can connect freely in an ad-hoc wireless network because there isn’t a central point. This word describes all the things that can send and receive data, called nodes [13]. They use a network called ad-hoc, which is made up of nodes. As a result, the next person agreed to be a “router,” meaning that data could move between devices without connecting. Many people think that ad-hoc networks have been around for a long time, but this isn’t true. In their plan, they want to build a communication ecosystem that grows without the help of any other infrastructure, just their phones [14]. This means a mobile phone can connect two phones to talk. Two people who can’t speak face-to-face can find a device that can act as a relay.

People have a lot of trouble because they don’t know how to make a network’s topology work the way it should. A person can compare two extremes and see how they compare. This network doesn’t have a set of rules [15]. It has long-range transmitters so that all nodes can talk to each other simultaneously. People will do more work overall because there may be many things they can use again. In the solution for ad-hoc networks made by the company, the IP environment is used to connect to networks.

All the mobiles that act as gateways, usually all mobiles, have routers in their networks to connect to the internet and other networks [16]. This means that most problems happen because of how the internet works. Networks that aren’t set up can be used in many different ways. They make it easy to build networks quickly, like in the case of an earthquake or a big meeting, so that people can connect soon. Wi-Fi is one of many wireless networks people can use [17].

A wireless ad hoc network is a collection of devices that creates a network on their own (WANET). Ad hoc is a term used to describe actions taken rapidly or only temporarily. One can make a wireless connection to another computer or device using ad hoc mode without utilizing a Wi-Fi access point or router. Ad hoc connections can also result in decentralized P2P networks, which don’t rely on a

previous network design. In contrast to traditional network architecture, where data continuously flows into and out of a central device like a router before it reaches the child devices, every node in an ad hoc network communicates with every other node in the same way (such as phones and PCs). Different wireless ad hoc network types:

A disaster rescues ad hoc network describes as a disaster that destroys the standard communication infrastructure; ad hoc networks are the only way to communicate information.

A mobile ad hoc network (MANET) is one where a mobile phone feeds the grid.

Army tactical MENT describes as due to its flexibility and quick deployment, the military uses wireless tactical ad hoc networks for on-the-go communication.

A wireless mesh network: Since nodes in a mesh network can communicate directly, this ad hoc networking is particularly beneficial.

A vehicular ad hoc network (VANET) facilitates vehicle communication. In intelligent VANETs, ad hoc networks and artificial intelligence collaborate to plan reaction actions in the case of an incident.

The wireless sensor network is described as wirelessly gathered data from noise, humidity, temperature, and pressure sensors that can function together in an unofficial network. As a result, routers are no longer required to establish a direct connection to the central hub each time devices need to convey information.

A smartphone ad hoc network (SPAN) is described as using Bluetooth and Wi-Fi; wireless ad hoc networks can be built between cell phones.

1.1. Motivation to study ad hoc network

Laptops and mobile phones that can access web services and make phone calls while moving are getting better at using technology that makes it easier to do these things. In the future, self-organizing networks will help cut the cost of communication by making it easier for people to connect [18]. To create and set up an ad hoc network. Ad-hoc networks are only suitable for people who need to move many files between devices. An “ad-hoc” network can be set up almost anywhere. So, businesses, organizations, and even general use at home may find them helpful. This makes it easier to use and is usually cheaper for a company [19]. It’s essential for people who study an ad-hoc wireless network to keep these things in mind, as shown in Fig. 3.

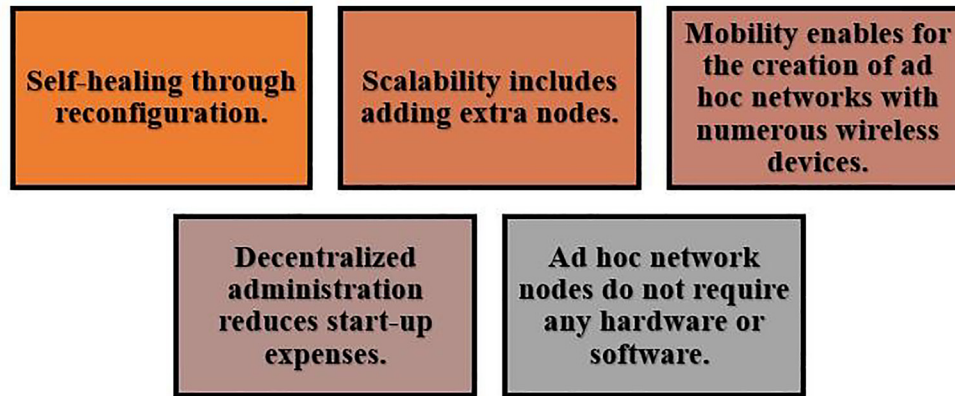


Fig. 3. Features of Wireless Ad Hoc Network.

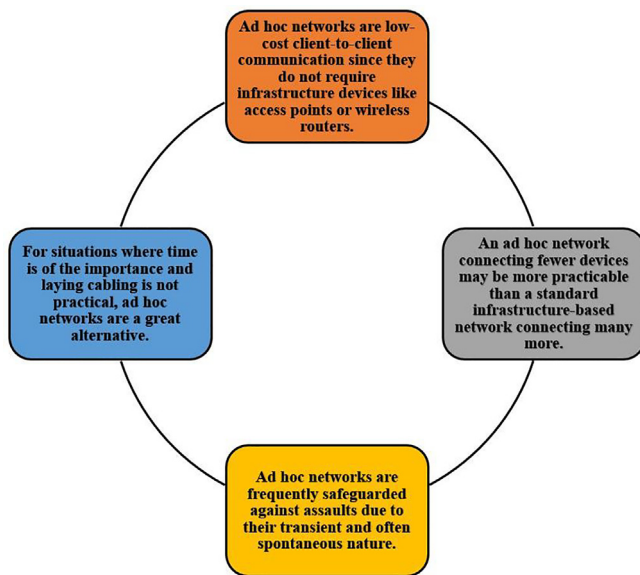


Fig. 4. Benefits of Wireless Ad Hoc Network.

Ad hoc mode can be easier to set up than infrastructure mode if users only need to connect two things. They do not need a router to set up a Wi-Fi network in a hotel room with no Wi-Fi. They can connect without one, and this mode is called ad-hoc. People can talk to each other over Wi-Fi signals without having an internet connection because the standard is based on ad hoc mode [20]. In this Fig. 4, readers can see that they also have other wireless ad hoc network benefits.

1.2. Vulnerabilities of ad hoc network

A network's structure is made up of itself. It also takes care of itself when things go wrong. It quickly forms when things change, and there isn't much electricity [21]. On the spot, there is no one to trust. As long as wireless devices can be used in an ad-hoc mode, some drawbacks must be considered before they are used. Because there isn't a way to track a signal's strength, there is no way to measure it. This isn't a big deal because many devices are always in range of each other. Network security is essential to consider as soon as all devices are linked. Users don't have much protection compared to the new Wi-Fi security features. The problems with some Wi-Fi technology are not very good at all.

Some Android phones and tablets can only connect to networks in internetwork mode, even if they can connect in an ad-hoc way [22]. Sometimes, third-party software can be installed on endpoint devices to communicate with each other on the spur of the moment. Ad hoc mode is best to set up a more extensive and permanent network to serve more people. Infrastructure mode is better than ad hoc mode. People who use access point wireless routers often have more powerful radio waves and antennas to cover a more extensive area. People who set up ad-hoc networks have trouble with wireless communication range because the antennas of endpoints don't have enough power. It can be hard to set up ad hoc wireless networks, like in the Fig. 5.

Ad hoc networks also don't work very well either [23]. Ad-hoc networks become more challenging to manage as the number of devices grows. People who use them find it more challenging to keep track of. No person or group is responsible for all the traffic in an ad-hoc network. Each device in an ad hoc network must establish a direct P2P connection with each other device instead of traveling through a single access point as in a hub-and-spoke architecture. This means each device must connect directly. More devices must be used to send data when a machine can't get close enough to the device it wants to connect to, so more must be used [24]. This slows down getting the data to the other device, taking longer. Wireless ad hoc networks have some problems; they are still suitable for many personal and business applications, even though they have a few flaws. In an ad-hoc network, connecting two or more devices is easy if the network managers know how a WANET works and if it suits their business needs. Ad-hoc networks don't have to be set up in a single place. Networks don't have to use any old equipment, like routers, access points, or wireless infrastructure networks [25].

When there is a problem with routing, each node in the network helps by passing data to other nodes. This means that the choice of which nodes to send data to changes based on the network connection at any given time. Ad-hoc networks can move data by flooding and traditional routes. Every-one on the network has the same power and can connect to any other device in the same link range. This is called an ad-hoc network, and it was made up on the spot [25]. A wireless sensor network comprises autonomous devices or nodes spread out all over. These nodes use sensors to monitor the temperature, sound, pressure, etc. In the next step, the two work to send their data to a control center over the network. Eventually, this led to wireless sensor networks that let sensors be put almost anywhere and send their data over long distances through wireless networks. New sensors have made this possible in the last few years because they are small and robust [26]. Fig. 6 shows the flaws in the wireless ad hoc network.

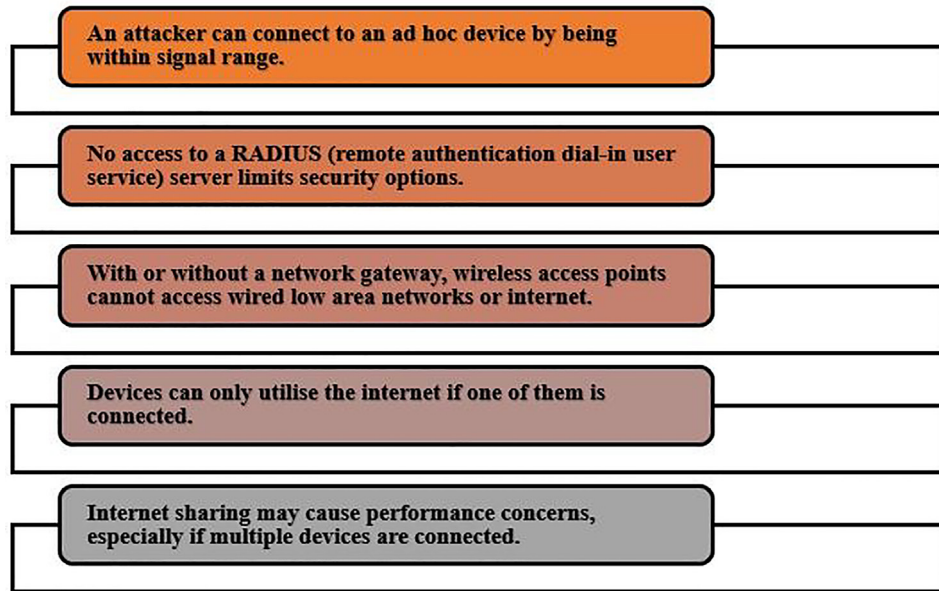


Fig. 5. Disadvantages of Wireless Ad Hoc Network.

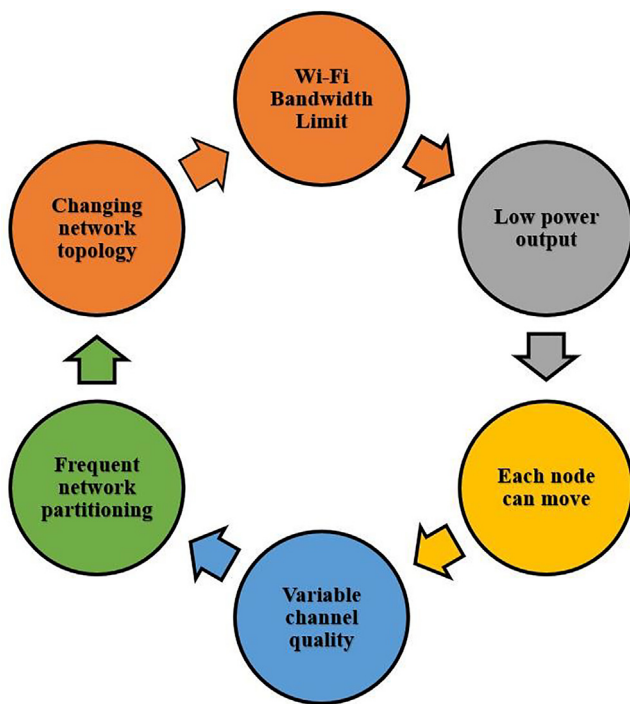


Fig. 6. Limitations of Wireless Ad Hoc Network.

1.3. Scope of this work

No person or group is in charge of the wireless ad hoc network because there is no fixed infrastructure or central management. People make it up as they go along, with no fixed infrastructure or central control. People who live next to each other in a wireless network can talk to each other if close enough. They can talk to each other even though they aren't in the same radio range. So, each one is also a wireless node. This means that even if one of the nodes moves or stops working, it can prevent the application from working, so this can happen. There are many parts to the endeavor that are shown in Fig. 7.

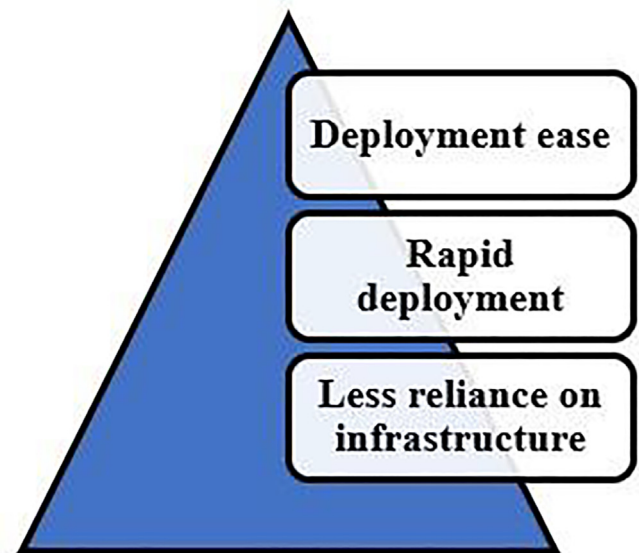


Fig. 7. Key Points of Scope of Work.

The best nodes are given more routing and computing tasks to solve this problem. They connect any two points that aren't next to each other. There is something that looks like a tree or ring. They are linked together in this way. Backbone nodes make up a virtual network that is made up of other nodes. Setting up and keeping track of a virtual network route is straightforward.

1.4. Research on ad hoc network

It is a part of a self-aware system that connects wirelessly. Ad-hoc networks don't have physical representations of the nodes, so they cannot see them. They don't use that representation to talk to each other. People in an ad-hoc group start communicating right away. Radio waves connect each node to other nodes in the network [27]. The network must have a fixed access point (AP) or base

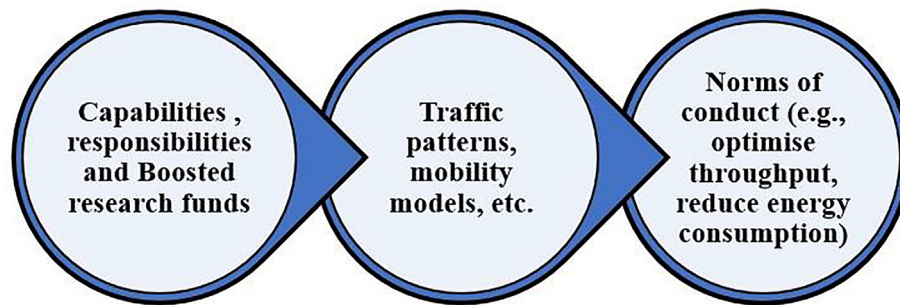


Fig. 8. Key points to research on wireless ad hoc network.

station to stay together. Fig. 8 shows what to look for when users study wireless ad hoc networks.

The web is spread out, and the nodes talk to each other. They are part of an ad hoc network, which means they are made up as they go along. Automatically, it links up with other parts of the network. Each node can set up its wireless network of other nodes that independently set up and manage the web. As time goes on, it will be a big part of pervasive computing because it's flexible, doesn't need a lot of infrastructures, and is easy to set up. It can also be used for many different things [28]. There are different types of wireless ad hoc networks and how they work. It also discusses some complex problems and comes up with a good solution based on new technology, which is what it does.

1.5. Organization of the paper

Paper described in nine different sub-sections: section 1 describes introduction related to ad hoc network, motivation to study ad hoc network, vulnerabilities of ad hoc network, the scope of this work, and research on ad hoc network, whereas section 2 describes significant findings in ad hoc network by different authors from 2000 to 2022 and section 3 illustrates types of ad hoc network such as mobile ad hoc network, smartphone ad hoc network, wireless mesh network and wireless sensor network. Section 4 describes mobile ad hoc networks, which are of two types: vehicular ad hoc networks and intelligent vehicular ad hoc networks. Section 5 describes wireless mesh networks in detail, and section 6 describes wireless sensor networks. Section 7 describes a discussion of the comparison table that addresses issues like limitations, challenges, characteristics, etc. Section 8 describes the future direction in ad hoc network research, and section 9 summarises the paper in the form of a conclusion.

2. Major findings

In an ad-hoc network, some devices can connect and communicate with each other when they need to. It usually doesn't last very long, as researchers learned at the start of our meeting. A network has been set up that allows people at her meetings to share information, even if they use different computers or phones [29]. There is no network hardware in ad-hoc networks, so there is no way to connect to them. In an ad-hoc network, all devices in a particular area can connect wirelessly, rather than having a fixed server or router act as a hub. When two people are near each other, they can connect without setting up a wireless access point. This type of network is called a distributed or ad hoc network. These are the things that let two or more wireless cards connect. People can connect wirelessly to a card called Wi-Fi [30]. They relate in a way that isn't meant to happen. Table 1 shows significant findings in ad hoc networks by different authors from 2000 to 2022.

3. Types of ad hoc network

Users don't need a router or wireless base station to connect to the internet. When using an ad hoc network, users don't need to use a router or wireless base station. A local area network (LAN) is called an ad hoc network. It is only used for a short time. Ad hoc networks can be set up for good, and when they are, they become wireless networks [61]. To describe a solution that is only for one thing, called an ad-hoc network, when people and things that want to communicate come together and talk to each other. Only for a short amount of time. The wireless card links both of them together [62]. The connection between the two devices will break down. Several devices can connect simultaneously to an ad hoc network, but the network may not work. Ad hoc mode is another way to say that a network is set up on the fly. There are different types of ad hoc networks, as shown in Fig. 9.

3.1. Mobile ad hoc networks (MANETs)

It is built on a link-layer network and a wireless ad hoc network. It also allows for a network that can be used to move around. As the nodes move, they connect wireless data. This group of nodes is called a network [63]. It is self-configuring and self-healing, and there is no setup. The network's structure changes, so MANET nodes can move where they want. These networks can be set up on the spot and use mobile devices (laptops, smartphones, or iPads) and wireless devices to connect and communicate [64]. Fig. 10 shows the features of MANETs.

Every time a node sends a message, it sends it to other nodes chosen to get it. There are two ways to use MANETs: on their own or as part of a more extensive internet. They can be used in both ways. Adding different transceivers between nodes makes it very dynamic and autonomous. Keeping traffic moving smoothly is the primary goal of MANET [65]. Each device must have the information it needs to keep things running smoothly. On the road, safety is a good use for sensors. Sensors can be used for the environment, home health, disaster rescue, air, land, navy defense and weaponry, and robotics. This leads to mobile ad hoc networks, or MANETs, set up on the fly.

3.2. Smart phone ad hoc networks (SPANs)

Sometimes when people use their phones, they are called ad hoc groups. If a group of phones is close, they can form an ad hoc network if they have ad hoc network technologies. People who use smartphones don't need to use traditional network infrastructure to build peer-to-peer networks [66]. They don't need cellular carriers, wireless access points, or traditional network infrastructure. Smartphones already on the market have Bluetooth and Wi-Fi technology built-in. Wi-Fi SPANs use the same idea as Wi-Fi ad-hoc mode, which lets phones talk to each other immedi-

Table 1
Major Findings in Ad Hoc Network by Different Authors from 2000 to 2022.

Year	Author and Reference	Major Findings
2000	Samir R. Das et al. [31]	Layers of routing and transport desktop and laptop personal computers with wireless low-area networks. Inadequate performance after two hops at medium–high loads.
2001	David B. Johnson et al. [32]	There are eight nodes in this wireless ad hoc network. It brings together mobile internet protocol and the Internet infrastructure. Network protocols must be used to solve future radio propagation problems.
2002	Mario Gerla et al. [33]	Wireless ad hoc network radio capability is the same. Ad-hoc networks cannot scale. This network is easily scalable as a large-scale ad-hoc network using several radios Initial multi-hop node clustering.
2003	Jetcheva, Jorjeta G., et al. [34]	There are few real-world mobile ad hoc network applications. Researchers give new movement data for large-scale ad hoc network modeling wireless ad hoc network routing for wide-area communication with multitiered design. The backbone network comprises wireless devices aboard buses and delivery vehicles.
2004	Han, Lu et al. [35]	Wireless devices form mobile ad-hoc network nodes within and beyond their radio range. A wireless ad-hoc network node can be a gateway. Almost all wireless nodes connect with base stations and access points rather than hop-by-hop forwarding.
2005	Chen, Jiancong, et al. [36]	Wireless networks are frequently used to deliver mobile internet access. The web is constrained. Both the access point and mobile nodes can be in ad hoc mode. This minimizes access point traffic and enhances network capacity.
2006	Rubinstein, Marcelo G., et al. [37]	Ad hoc networks are great for military, emergency, and disaster recovery. A growing mobile market needs smooth internet working. Ad hoc commercial use of new wireless technologies an ad hoc network must feature noisy, short-range, unsecured wireless transmissions.
2007	Don Towsley et al. [38]	With a wired base station on top, it can handle ad hoc networks. This network design is used for a cellular network with multiple stops. These models figure out how different network dimensions and forms can have an effect.
2008	Mahfoudh, Saoucene et al. [39]	Since energy is limited, networks must be set up to last as long as possible. These include reducing the number of data transfers and planning when nodes will sleep. Each location has unique approaches. Mixing solutions from different fields make the most of energy efficiency using all four methods.
2009	Martin Haenggi et al. [40]	Interference slows down wireless ad hoc networks. The outages are temporally and geographically correlated due to interference in a wireless ad hoc network with nodes distributed over an aircraft.
2010	Leandros Tassiulas et al. [41]	Ad hoc networking protocols use social network analysis concepts like centrality measurements and community formation to measure the network. When communication is not planned, it is harder to use other systematic methods like cross-layer optimization.
2011	Bakht, Humayun, et al. [42]	Routing is complex in mobile ad-hoc networks. It will aid in better understanding the issue area and can be used to improve or extend existing approaches.
2012	Biswanath Mukherjee et al. [43]	The end-user obtains freedom from tethering and saves money. Ad hoc networks are controlled by users, whereas authorities control infrastructure networks. An ad hoc network has several hops to attain its goal. Wireless routers strategically positioned around the web allow multi-hopping.
2013	Renu Dhir et al. [44]	Ad hoc networks have dynamic mobility because nodes can join or leave the network. Nodes are network nodes. They can be hosts or routers. Nodes in ad hoc networks must be secure and respond quickly (end-to-end, intermediate, and wireless).
2014	Al-Sultan, Saif, et al. [45]	Vehicular ad hoc networks can assist in enhancing traffic flow and passenger comfort. Vehicular ad hoc network has grown in popularity among scholars investigating wireless and mobile communications.
2015	Mohammed Aljohani et al. [46]	An ad hoc network provides decentralized device access. Smartphones running android may transfer data through ad hoc networks. Wi-Fi devices can communicate faster and further than Bluetooth ones. Ad hoc android hardware mode. Without an access point, android devices may communicate directly.
2016	T. Shanthi et al. [47]	It consists of wireless mobile nodes (devices) that may build a temporary network without using existing network infrastructure or centralized administration.
2017	Giorgio Quer et al. [48]	So long as the target can be reached directly or with minimum retransmissions via peers, device-to-device solutions look promising. In this way, cellular networks may free up local traffic. Control and data streams are separated in software-defined networks, and the cellular base station can help coordinate the ad hoc network.
2018	Tkachov, Vitalii, et al. [49]	Border nodes limit data transmission speeds in wireless sensor networks. Sensors act as routers because data flows are constantly rerouted. A strong electromagnetic field boosts energy cost. It is undesirable in highly specialized hidden sensor networks.
2019	T. Gowthaman et al. [50]	After a natural disaster, save our soul's wireless ad hoc network technology can be utilized to re-establish connectivity. Communication with individuals in need is helpful. This network is easily accessible to victims in need. The network will send a single word. RF transmissions replace cable and mobile communication.
2020	Alaa Hamid Mohammed et al. [51]	Mobile ad hoc wireless networks create ubiquitous networks with no central administration or infrastructure. The mobility of the intermediate node between the source and destination creates an unstable topology. The wireless design relies on route routing and protocol selection.
2021	Abhishek Majumder et al. [52]	A novel route selection strategy combines ad hoc on-demand distance vector and ant colony optimization to boost the quality of service in mobile ad hoc networks. The path's pheromone value determines the ideal data transmission route. The data packet will be transmitted via the most pheromone-rich way.
2022	Afreen Khurshed et al. [53]	Heterogenous ad hoc network mobile devices like phones, laptops, tablets, and MacBooks have wireless transceivers and computational power. It acts as a router for data transfer and distant communication. Ad-hoc networks have mobility management, battery management, route finding, and maintenance challenges.
2022	S. Indira Gandhi et al. [54]	In-flight Wi-Fi links devices to the ground, enabling the internet of things. Remote or maritime areas employ aeronautical ad hoc networks. A quick analysis is required to build a realistic mobility model and network routing with high quality of service. Improving the quality-of-service data communication is challenging.
2022	Xin Tong et al. [55]	Clustering improves the scalability of planar ad hoc networks. Overhead is a critical performance factor when clustering a planar ad hoc network. The overhead performance of planar and clustered networks may be compared by combining their routing, scheduling, and cluster head selection techniques.
2022	Elaryh Makki et al. [56]	Voice over internet protocol performs well in vehicular ad hoc networks. These networks have several issues supporting voice conversations with appropriate quality of service. Optimized link-state routing protocol tests voice over internet protocol performance on the vehicular ad hoc network.
2022	Arvind Kumar Shukla et al. [57]	The new method improves and takes care of the network. To find a way to deal with the problem of moving nodes while routing.
2022	Alzahrani, Eman, et al. [58]	Media access control protocol regulates node access to the shared wireless medium in energy-constrained networks. A suitable media access control protocol eliminates overhearing, collisions, and idle listening, extending network longevity. Designing a media access control protocol that employs quorum to decrease idle listening and overhearing is economical.
2022	Katiyar, Abhay, et al. [59]	Clustering addresses vehicular ad hoc network fast topological change, improving network stability and scalability. With multi-hop clustering, vehicular ad hoc network overcomes poor cluster coverage, excessive cluster formation rates, and inefficient network data processing.

Table 1 (continued)

Year	Author and Reference	Major Findings
2022	Wheeb, Ali H., et al. [60]	Uncrewed aerial vehicles have recently gained popularity for consumer, commercial, and military use. Flying ad hoc networks are frequently deployed in three dimensions, with a mobility model determined by their work, which differs from vehicular ad hoc networks and mobile ad hoc networks. Designing routing protocols for flying ad hoc networks is also tricky because of their dynamic topology and flight constraints.

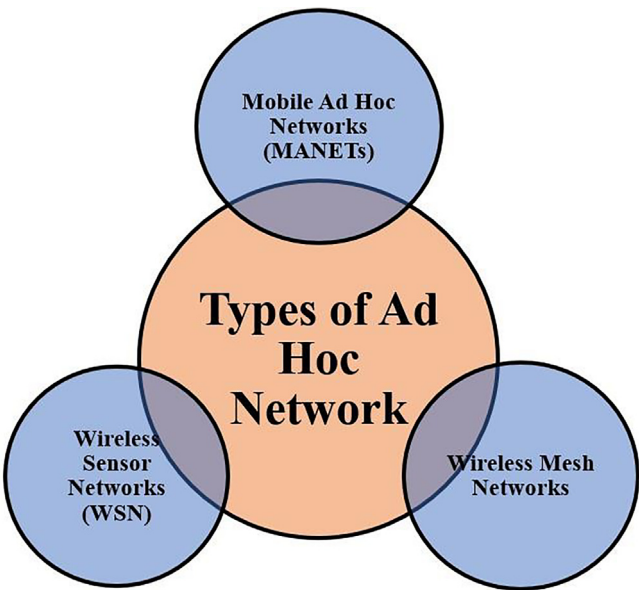


Fig. 9. Types of Ad Hoc Network.

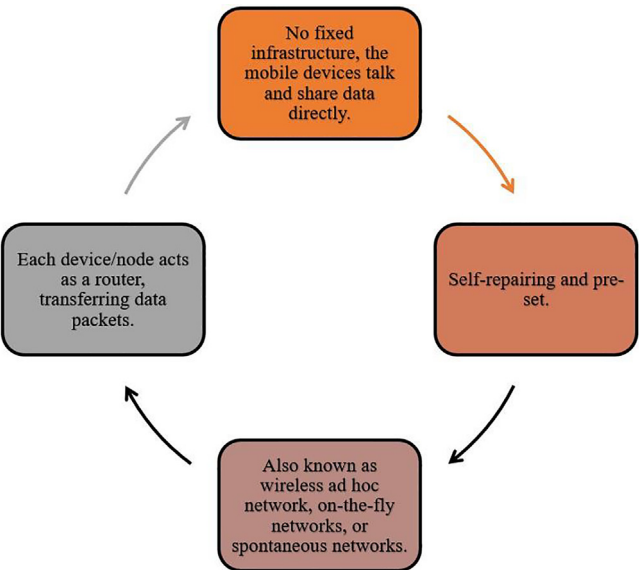


Fig. 10. MANETs Features.

ately through a transparent neighbor and route-finding method called fine neighbor. Multi-hop routing (ad hoc routing) and relays made SPANs different from the standard hub and spoke networks like Wi-Fi direct. There is no group leader, so peers can join and leave the web, and there is no way to know who is in charge of the group [67]. SPANs don't have a group leader, too. It's going quickly, making it easier for people to talk to each other and con-

nect. They need a unique infrastructure to connect the wireless devices, so these intelligent devices need that.

In some cases, people can't rely on the telecom infrastructure. High costs or not enough infrastructure could be why this isn't possible. So, think about how infrastructure-based networks would work if there were storms or earthquakes [68]. These disasters could destroy the base station and phone lines. It can be hard to communicate in war zones because there may not be enough infrastructure for people to use, like army training grounds. People who live in cities with many cars use a lot of internet maps because they can find the least crowded routes. Users need to have networks built into the ground to use online maps [69–71]. It's possible to use MANET, which lets mobile devices connect without setting up any network infrastructure. The data has to go through many other things to reach its destination. So, each MANET device can send and receive information and be a sender and router. Because the Android operating system can be changed to meet the needs of different people, it can be used by every-one [72]. One of them is called the optimized link-state routing protocol.

Phones, routers, and laptops all work better with it now that it has been made to run faster [73,74]. There is a name for networks like this one. They are called mesh networks. For even more stats, they have come up with a new name for it reachable. Flexible Routing uses this measure to show how much bandwidth, end-to-end delay, and hops are in a mesh network. This is a new way to show these things. This is a measure that is used in Flexible Routes. Fig. 11 shows some of the elements that SPANs have. It can also be quickly and simply shared.

A mesh network of smartphones is formed that hasn't been set up, so it doesn't work well. In this case, these phones can talk to each other even without the help of a service provider network. To use the phone, it would need to be changed so that the standard security restrictions could be turned off so that the phone could be used [75]. Android phones have this software to make voice calls, send text messages, and send files to each other. Google Android phones don't have to be in the same price range as other phones. People who need the data will get it as long as other Android phones in the area have the same configuration [76]. There is a way for people to share internet access and use programs set up in their own homes. There are a lot of wireless devices that this software can work with. These include smartphones and laptops, desktop computers, and wireless routers, and it works with them all. It uses the OLSR routing protocol to talk to each other inside, called W-Fi direct [77].

Whenever two or more things come together, they can share information. All of the devices must be able to use Wi-Fi direct. Because it can only be used between two people, there is only one problem with it as long as users have a Bluetooth device, users can set up ad-hoc networks to send and receive data over short distances and low ranges [78]. People in Bluetooth have a master and an enslaved person who must work together. The topology is like a piece of cake. There are seven enslaved people that the master can connect to simultaneously, but only one sends data to the master. They do this for all the enslaved people linked to the master in this way [79].

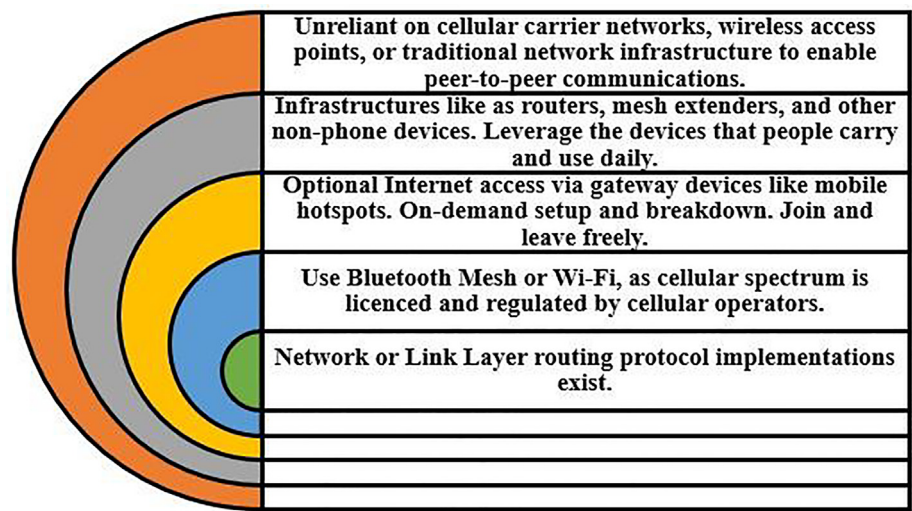


Fig. 11. Features of SPANs.

3.3. Wireless mesh network

Multi-hop wireless networks are made up of many wireless mesh routers [80,81]. A wireless mesh network is called a multi-hop wireless network. These routers are all connected wirelessly by a mesh-like backbone that runs through them. A laptop or a smartphone with wireless connectivity lets users connect to the network through a router that acts as a wireless access point and links to the router. People use the backbone mesh network to send and get data from and to each other, and they do this through it [82]. Wireless networks connect to one or more routers, which relate to different networks, like the internet. The following things are done with wireless mesh networks, also called wireless ad hoc networks (WANET), as shown in Fig. 12.

Two or more wireless radios work together to build an RF channel that connects all of them to talk to each other. They use routing protocols to combine all of the radios. No matter how many radios the system has, it has one single name identification (SSID) and one IP address for the whole thing, making it easy to tell it apart from other wireless or mesh networks. This makes it easy to tell it apart from other networks [83,84]. They connect these nodes, called a

wireless access point (WAP), to form a network. A wireless mesh network (WMN) has wireless access point nodes. Each node only has to send as far as the next one to set up a network.

3.4. Wireless sensor networks (WSNs)

Wireless Sensor Network (WSN) is a network that doesn't need any kind of infrastructure. Users can see how the system works, the world around them, and more wireless sensors [85,86]. People use WSNs to manage and monitor the environment in certain places. WSN system's base station is connected to the internet to share data with other people. Sensor nodes that communicate with each other wirelessly are called WSNs. These people use this information to discover more about their lives [87]. Nodes are often low-power and spread out in a way that doesn't make sense. When it comes to setting up security for WSNs, there are a lot of limitations. They have limited memory, compute power, battery life, and bandwidth, making it hard to set up security [88,89]. A person can be attacked by someone who wants to invade their privacy, control, or availability. It also advises on how to keep the keys safe. They discuss a new routing method for WSNs that considers their energy

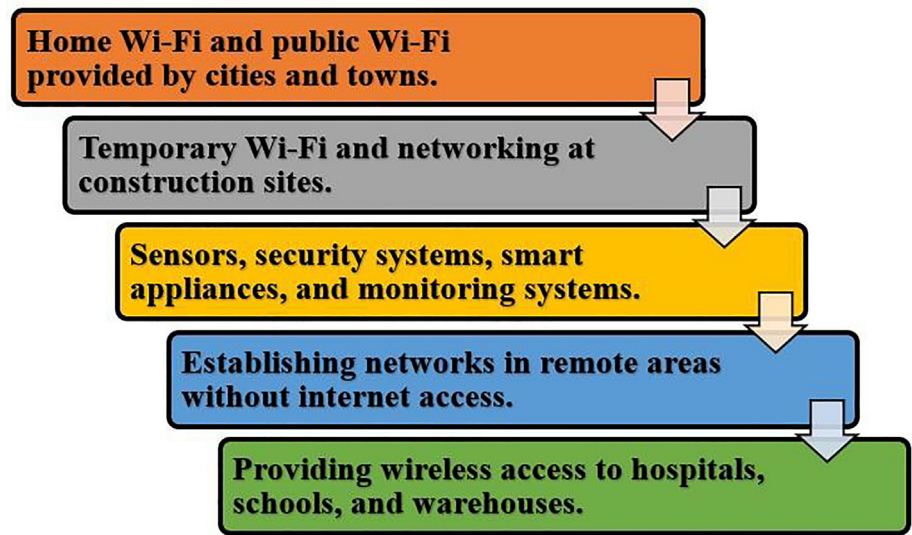


Fig. 12. WMN are frequently used for the following purpose.

usage. A lot of critical technology is used to make the Internet of Things (IoT) work, and then it talks about WSN, which helps make that happen. To keep IoT safe, there will be a lot of devices, and the system is open [90]. People who use wireless sensor networks can see parts together in Fig. 13.

4. Mobile ad hoc networks (MANETs)

People are moving away from traditional desktop computing, where workstations connect through shared servers on a single network, to an environment where many different platforms communicate over various networks [91,92]. This is called multiple network communication. This change is taking place very quickly. It changes and adapts to meet the needs of workers on the move and their work teams. There will be a lot of independent mobile users in the next generation of wireless communication systems, so they will need to be added quickly. MANETs all nodes in a network architecture talk to each other without having a single person in charge. Instead, each node could act as a router.

MANETs are good because they can be changed and set up quickly. It's still researched into MANETs, but they aren't used very often [93]. People need to use the correct routing protocols to communicate quickly. Unique routing methods need to be used to deal with the changing topology. People who use flat routing methods may use them on small networks.

On the other hand, people who have more extensive networks will need to use either hierarchical or global routing to get their data. Because there is no single protocol that works with all networks, there isn't one that works [94–96]. Choosing the protocols is essential based on how many nodes there are, how big it is, and how quickly the nodes move. At the same time as wireless networking, ad hoc networks can be exciting and hard to set up and break down. When there is a network, some nodes can be moved. Because the network topology can change quickly and without warning, the web can be very different [97]. They are more flexible and cheaper than traditional networks, consisting of devices to form networks independently. The working of MANETs is shown in Fig. 14.

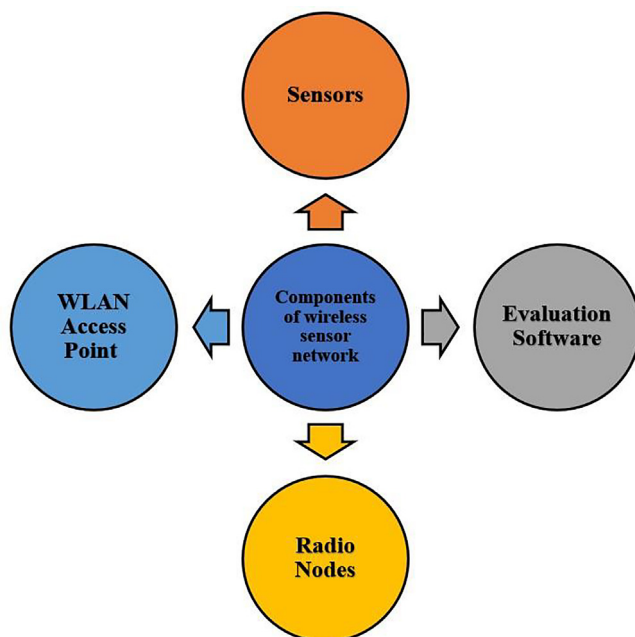


Fig. 13. Components of wireless sensor network.

This means they have to deal with the effects of radio transmissions like noise and interference. Because of this, the links on a wired network don't all have the same bandwidth. Host and Router are two things that each node does. This means that each node has a say in how things work. This is because nodes can leave, new nodes can be added, and there is a chance that mobile nodes could move around, which means that the network topology changes over time [98].

4.1. Types of MANETS

Self-configuring networks are called mobile ad-hoc networks. These are networks made up of mobile devices that connect through wireless links. A MANET device will change its relations with other devices when it moves, so this is how it will work when it does this [99,100]. People who use a MANET can move in any direction. Each must be a router because it must drive traffic that isn't for its use. MANETs are made by giving each device the information to direct traffic in the right direction. These networks can be separate from the more considerable internet or linked [101] Fig. 15 shows how VANET and MANET differ for different parameters.

4.1.1. Vehicular ad hoc network (VANET)

Vehicle-to-vehicle networks, or VANETs, are built as they go along. They are made up of different vehicles and other things that connect them, like bridges. They communicate wireless and share essential things [102,103]. Also, a small network of cars and other devices like phones and TVs act as nodes. Wireless technologies like cell phones, satellites, and WiMAX are also called wireless, and factors that describe VANET are shown in Fig. 16. There are wireless networks made up of cars and other vehicles that use the principles of MANETs. These networks are called vehicular ad hoc networks (VANETs). When VANETs was first discussed and made known in 2001, they connected cars and shared information [104]. There are ways to build networks, and cars can share information with other vehicles. In the intelligent transportation systems (ITS) framework, VANETs are part of it. One of the other names for this network is the intelligent transportation network or VANET.

They are thought to have grown into a more extensive network of cars, likely to develop into a network of self-driving cars soon. VANET has to change its topology quickly, unlike other wireless networks, because its nodes move so much. The number of cars with computers and wireless communication devices is growing [105].

There is a lot to talk about in the field of intervehicle communication. Many different things can be done with VANETs, like making sure people don't get hit by cars, avoiding blind crossings, etc. People in developed countries are starting to use VANET more and more to control traffic during rush hour, which is good. It can save many lives daily in well-being and non-security applications [106]. VANET uses special steering conventions to ensure that the source and the target are connected. VANET moves around a lot and has network problems, so it is hard to figure out a solid end-to-end path between them. VANET's biggest problem isn't how to steer but how to work together as hubs, no matter what. Even the best guidebook would be pointless if the seats didn't help spread the word about what they were telling people. Besides that, this explains how mobile social networking can speak about how people work together in groups. The idea of mobile social networking could be used with traditional VANET collaboration tools so more people can help at a hub [107]. VANETs are very important because they protect people by reducing congestion and giving ambulances and other vehicles a safe way to get around things, as shown in Fig. 17.

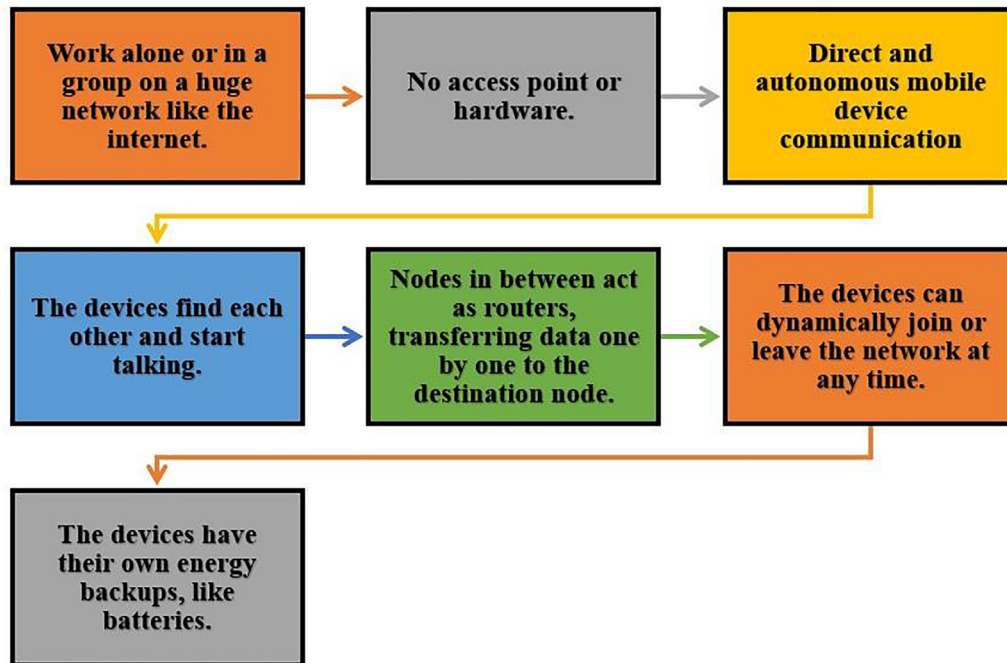


Fig. 14. Steps of MANET Working.

Such networks connect moving cars in a particular area, vehicle-to-vehicle (V2V) communication when a car talks to another car immediately. This case sends the nodes' information to their other nodes [108,109]. Some nodes work to make sense of this information and then send it to others. People can join and leave a network when they want to, making it more flexible and easier to use. VANETs will use technology like Dedicated Short-Range Communications Wi-Fi to connect cars. There is a type of communication called vehicle-to-infrastructure (V2I). This is when a vehicle talks to a piece of infrastructure, like a roadside unit. Because of a network called VANET, people from all walks of life have become interested in IT [110,111].

There are many applications for ITS that use this network as the primary source of power. There is a good chance that VANET could make roads safer, traffic flow better, and commuters more comfortable. In the case of VANET, a type of network that comes from building an entire network of cars for one specific need or situation, this network is called VANET and is made up on the spot. It can predict and help them with road safety and other essential things [112,113]. People are more interested in VANETs now because they can be used in many ways. Fig. 18 shows different international groups that work to make VANETs more standard.

VANET combines the abilities of new wireless networks and the abilities of cars to make it even better. Two or more cars talk to each other in the same area. Car and other vehicles share much information [114,115]. This includes safety information, traffic jams, traveler information, and entertainment. It makes it easier for every-one to get where they need to go. VANET apps are divided into five groups: safety-oriented, non-safety-oriented, commercial, convenient, and productive. Fig. 19 shows the security services that can be used in VANET.

A network called VANET is always moving because cars are nodes in the network. This makes the network always moving. Mobile internet is formed when cars connect and lose signal range [116,117]. Police and fire trucks will talk to each other at first to keep people safe, as shown in Fig. 20 how the VANET could go wrong.

It can let cars and roads talk about dangerous situations, making commuters less stressed. People now know that VANETs, which cars use on highways and in cities to talk, work well, and VANET is a subclass of mobile ad hoc networks [118,119]. It has the power to make life or death decisions for drivers and other people who use them. They are exciting because they are very flexible, cheap, have suitable sensors, and handle problems. There are now many new and different ways to do things with remote sensing. People can now watch the world around them with wireless sensors that can sense, analyze, and send information. After each node sends its own data set, each node gets the data other nodes send to work together. Many new cars on the market now have sensors that enable them to join and merge into the network, which lets them use VANET's advantages [120–122].

4.1.1.1. Concept of clustering in VANET. Clustering in VANET is crucial to understanding clustering, which occurs when a group of cars shares the same characteristics. The most common VANET clustering strategies are mobility-focused, but many VANET clustering algorithms also use these strategies. A group is formed when two or more cars near each other have some features in joint [123]. In data mining and machine learning, grouping things together is often used. The same way clustering is used in VANET can also be used in MANET, the predecessor of VANET. It's called a "network" when many small networks or clusters work together to make more extensive networks, also called "networks."

Clustering in VANET has a "cluster head (CH)" process that helps it work. During the procedure, this is very important. There are many ways to make a cluster based on the input metrics. The cars comprise a cluster that is part of its cluster member (CM). There are also algorithms called cluster gateways (CGs) that use two CMs to communicate on behalf of the CH so that the CH can talk to other clusters [124]. Besides these two things, the CH and CM comprise the rest of the cluster. A cluster is made up of all the people who live together. They're all called CMs unless they aren't already. This is how a cluster is made up of many things: CH, CG, and CMs are all possible. CM is a node that moves around,

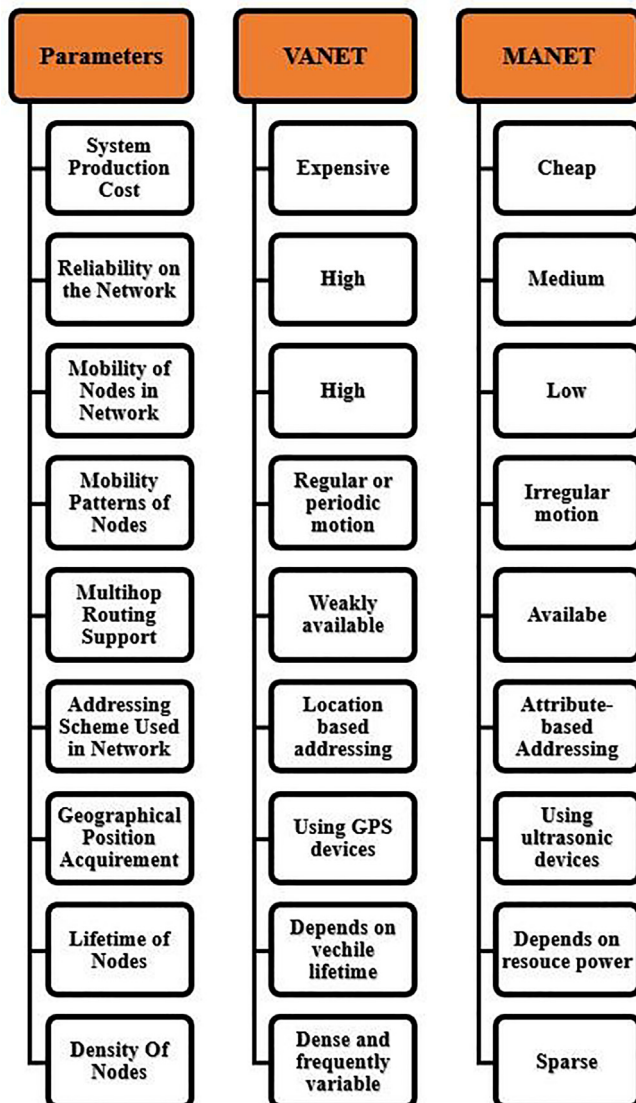


Fig. 15. Difference Between VANET and MANET.

and CH is a router that moves around. As long as CH and CM are everywhere, there's CG. To figure out which cars are in the cluster, researchers look at the vehicle's average relative velocity and acceleration, position and direction, vehicle degree, density, transmission range, and so on to figure out which cars are in the cluster [125]. Choose CH from vehicles that other people drive that are the most stable. It all comes together as CMs. Choosing a cluster member is part of making a cluster, and there are no extra rules for selecting a cluster member. It doesn't keep a list of routes for other clusters, which the CH does if it's essential. Large networks are considered small networks or clusters, so this is how they are thought of in the end [126].

CH can only go so far before it has to stop. People who drive near the edge of an object cluster are more likely to lose contact with the CH because they can only go so far before they run out of space [127]. People who drive fast can change the speeds of two cars. A lot of data will be lost when a vehicle stays close to the edge. One way to think about a cluster of cars is that they are 1-hop clusters. This means a CH can reach all its CMs directly because they are all in the CH range. When a vehicle can't get to the cluster's CH but can get to one of its members, it happens, too [128]. The new vehicle joins the group through one of its CMs. This is called "multi-hop clustering."

In machine learning and data mining, many ways have been found to make clustering as fast as possible. Clustering systems aren't the only way to make groups more efficient and stable. There are hybrid architectures for VANET clustering that uses machine learning and fuzzy logic to make the groups more efficient and stable by picking a better vehicle to be the CH [129]. VANET clustering surveys didn't look at innovative and fuzzy logic-based programmes like machine learning. In addition to intelligence-based strategies, many algorithms focus on how quickly things move to keep clusters stable. Cars and trucks also use network mobility (NEMO), similar to clustering. The mobile router (MR) moves from one place to another with its mobile nodes (MNs), and only the MR can communicate with the MNs. "CH" and "CMs" are two parts of the clustering algorithm [130]. Here, researchers can compare this situation to that of others. NEMO's MR and MN are the same as VANET's CH, CM, and RSU. VANET clustering strategies also allow the CH to cover more than one hop area to cut down on the number of groups.

4.1.2. Intelligent vehicular ad hoc network

Intelligent vehicular ad hoc networks (InVANETs) aren't going to replace cell phones as a way to communicate. There are ways to look at car information on electronic maps through the internet or special software, like Google maps. The good thing about a WiFi-based navigation system is that it can find one's car even in places like universities and airports that are very big, like cities. Use InVANET to find the best way to get where users need to go with the least number of people around [131]. The method can help people find and recognize things in new places as a city guide. There are intelligent transportation systems, which are made up of cars that can talk to each other. They can talk to each other and roadside access points, also known as Road Side Units (RSUs). Drivers will get all the information they need from vehicular communication at the right time, making roads safer, more efficient, and easier to travel. Communication between V2V and V2R is good because V2R is better for sparse networks and has long-distance communication. At the same time, V2V allows direct contact across short to medium distances and in areas with no access points on the road [132,133]. People can be safer and more comfortable if cars and trucks talk to each other and the ground. It might be good for cars to communicate in ad hoc networks with routing based on where they are. People use mobile IPv6 to keep a session going and reach the internet on mobile devices that need to use the internet. Solutions for mobile IPv6 in non-mobile networks already have it built-in, so it's already possible. Direct in-vehicle, vehicle-vehicle, and auto-roadside communication can all be used in a solution that uses a Mobile IPv6 proxy-based architecture. When more than one communication mode is used, the solution switches back and forth between vehicle-vehicle and roadside communication modes [134,135].

4.2. Routing protocols of MANET

People use protocols to make sure that any machine that is connected to them can read the data. A routing protocol is a set of rules that can be found and shared with the help of a MANET; there are numerous ways for people to find each other [136]. The three types of routing protocols are proactive, reactive, and hybrid, and they all work together. They are made to be able to handle a lot of nodes with very little space and resources. This is a big problem for routing protocols because nodes can go away and return. People should try to send messages faster even though there are more and more mobile nodes, even though that will take more and more time [137]. This is another thing you should think about. To connect, users can move the nodes around and connect them through a network set up on their own by the nodes. They can move around and

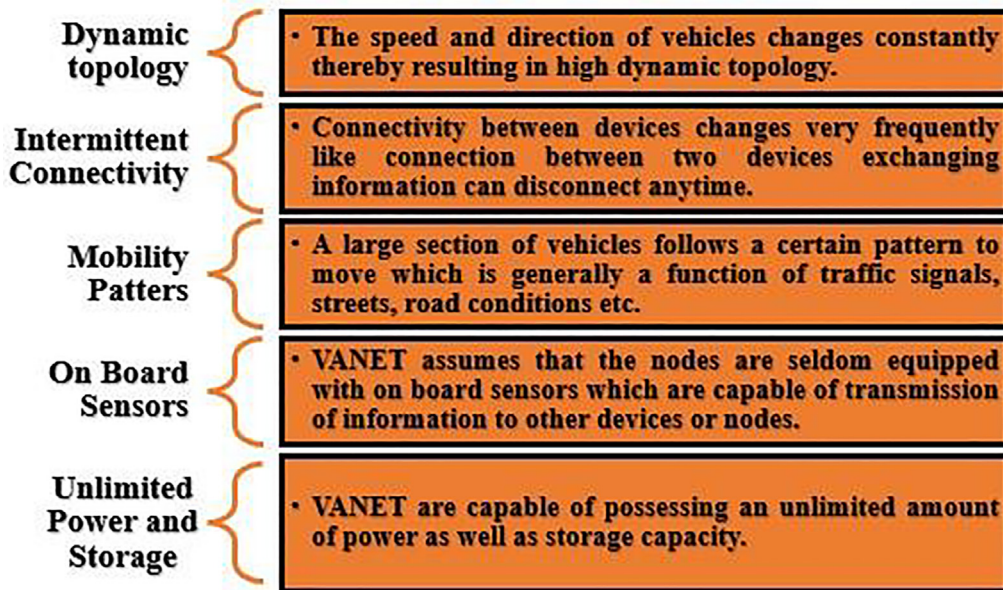


Fig. 16. Factor that can describe VANET.

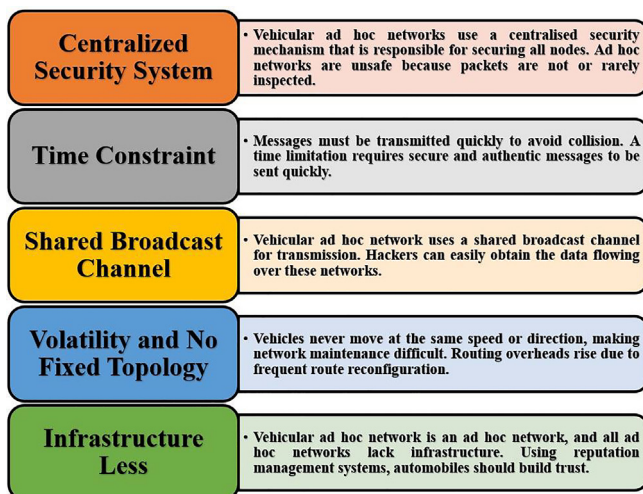


Fig. 17. VANET Features.

connect to other nodes at any time. Mobile nodes with limited bandwidth, dynamic topologies, limited energy, and infrastructure and link capacity changes. The routes between nodes give out information that lets two nodes on a network choose their routes [138]. Fig. 21 shows the different types of routing protocols that are used in the VANET, which is shown.

4.2.1. Proactive routing protocols

Link-state routing algorithms are used in this type of routing protocol, and they are used in this type of routing protocol [139]. They send out information about nearby links and do it all the time. They do this by exchanging control messages with their neighbors, which keeps their information about where to go up to date and saves them both time and money.

4.2.2. Reactive routing protocols

Proactive routing protocols have extra costs that reactive routing protocols don't, but they don't. A distance-vector routing algorithm is used to find routes [140]. It only builds a route to a specific

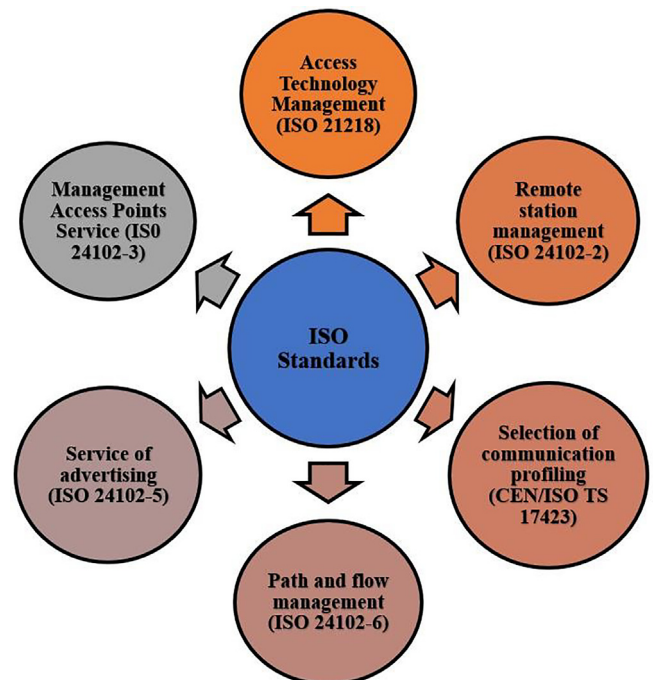


Fig. 18. VANETs ISO Standards.

destination when a node starts looking for routes and asks for them to be built. In MANET, there are a lot of very reactive protocols.

4.2.3. Hybrid routing protocols

It is a mixture of reactive and proactive routing protocols [141]. Table 2 show the performance matric of different routing protocols. Table 2: LSP stands for Local Shortest Path, N stands for No, M stands for Medium, Y stands for Yes, SP stands for Shortest Path, RT stands for Route Table, P stands for Possible, H stands for High, L stands for Low, A stands for Associativity, RC stands for Route Cache, and VH stands for Very High.

Proactive Routing Protocol – DSDV, OLSR, OSPF, and FSR.

Reactive Routing Protocol – AODV and DSR.

Hybrid Routing Protocol – TORA and ZRP.

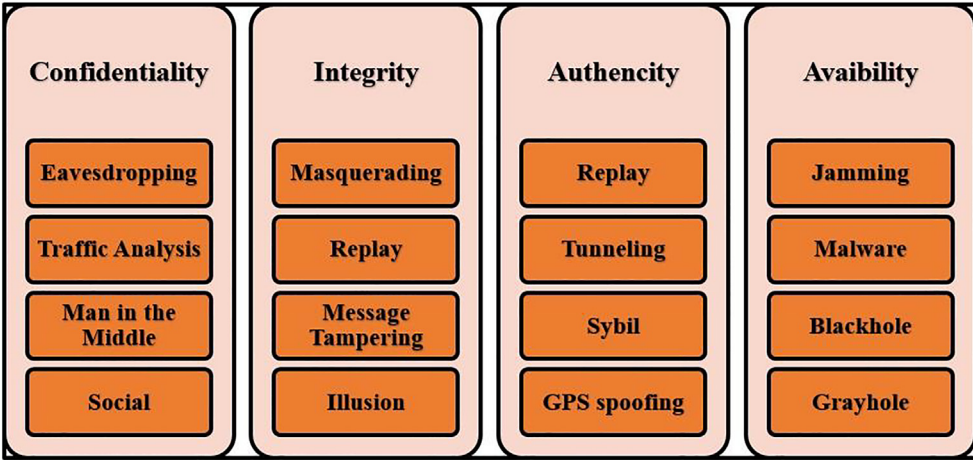


Fig. 19. VANETs Security Services.

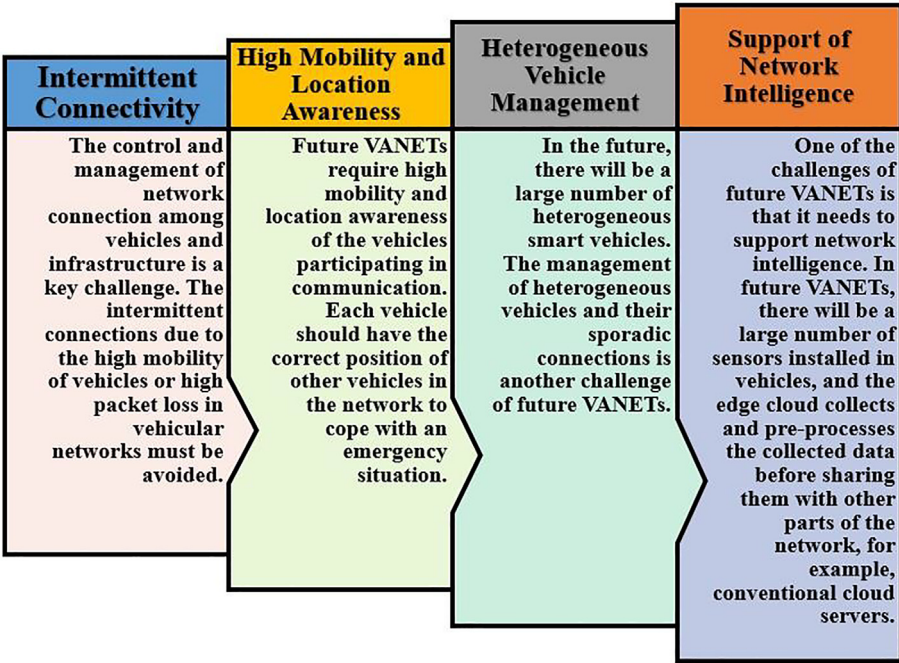


Fig. 20. VANET Future Challenges.

4.2.4. Round robin protocol

Over the last decade, people have been more likely to install networked systems because they can be used in many real-world situations, like communication and monitoring their health. Activators, controllers, filters, and sensors are just parts of a networked system that can send data across a shared channel [143]. This isn't always the case, though. This, on the other hand, isn't true. It can slow down communication when there isn't enough space for all the parts to send data simultaneously, which happens more often when there are a lot of parts. To keep data from getting mixed up, communication protocols have been used. They only allow a single thing to join the network at a time [144]. To communicate with other system parts quickly, users can use the Round-Robin protocol (RRP), which sends data between them in a set circular order. The Round-Robin protocol is used in networked control systems to send information. All the nodes in the network can connect individually at a set time [145]. This way, the network can save a lot of bandwidth. This protocol has been looked at from the per-

spective of hybrid systems. Time-delay systems were used to look at exponential stability and the features of networked control systems with Round Robin scheduling. A delay switching system model was used to look at these things. To know what other people in one's neighborhood are doing, users must send information to all the other nodes in their area. Users only listen to one at a time, following the Round Robin rule [146].

4.2.5. Try once discard protocol

As soon as users set a transmission order for nodes, they think about how to plan for network resources, and it's called "Try-Once-Discard." According to how much traffic there is, the Maximum-Error First (MEF) scheduling policy is used to move network resources around [147]. As for this kind of scheduling, priority scheduling is the term used to describe this. The process is called "TOD" because new information is used to make scheduling decisions. It's easy to use these protocols in a Lyapunov-based design because Lyapunov functions can be used to describe them [148].

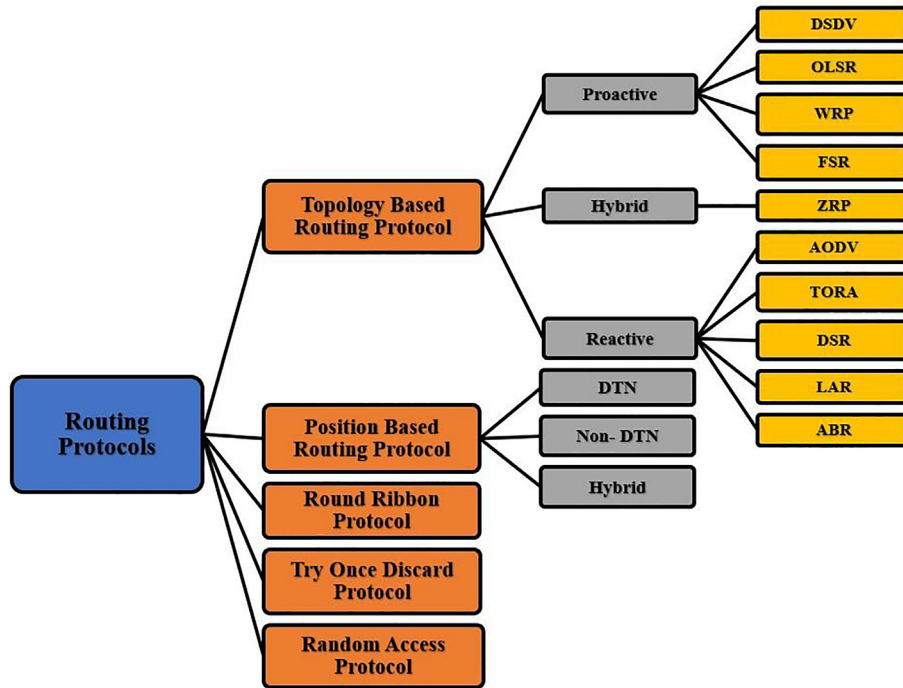


Fig. 21. MANET Routing Protocols.

Table 2
Performance matrix of different routing protocols [142].

Parameter	ZRP	AODV	TORA	LAR	ABR	DSR	DSDV	OLSR	WRP
Routing Metric	LSP	SP	SP	SP	A and SP	SP	SP	SP	SP
Use Sequence Number	N	Y	N	N	N	N	Y	Y	Y
QoS Support	N	N	N	N	N	N	N	Y	N
Routes Maintain in	Intrazone and Interzone Tables	RT	RT	RT	RT	RC	RT	RT	RC
Multicast Capability	Partly	Y	N	Y	N	N	N	Y	N
Congestion	M	M	L	M	M	L	H	M	M
Periodic Broadcast	N	P	P	P	–	N	Y	Y	–
Security	N	N	N	N	N	N	N	N	N
Error Message	M	More	–	Less	–	Less	More	–	Less
Hop Count	M	Normal	M	H	Less	VH	M	Less	M
Routing Over Head	M	H	H	L	L	L	M	M	H
Multiple Route	N	N	N	Y	N	Y	N	N	Y
Loop-Free	Y	Y	N	Y	Y	Y	Y	Y	Y

There are some rules about how people should talk to each other at work for TOD:

- Time slots have a set length, and all nodes must always be the same length.
- During each time slot, all of the nodes fight for the medium. Medium goes to the node with the most significant weighted error from the previously reported value, so that's the one.
- The win sends the current value it has to the destination node in this time slot to finish it off.
- For the TOD round to finish on time and the system to stay stable, an upper bound must be set for the combined length of network-wide value-based medium arbitration and message transfer.

The Controller Area Network (CAN) protocol, used in the car industry, is the best way to talk to TOD. The CAN system uses a priority-based system to ensure that arbitration times are always the same with no destructive collisions [149]. Because CAN was made to work only on wired, single-hop networks, it can only be

used that way. When TOD first came out, it wasn't clear if it could be used in wireless networks where more than one person used it. People can't use TOD immediately with wireless networks because they haven't used it yet. There is no way to get around TOD regarding wireless channels. Protocols are used in wireless single-hop networks to announce when a TOD has been reached. Researchers don't know any examples of wireless multi-hop networks that have worked in real life [150]. Users can use the Wireless Dominance (WiDom) protocol on wireless networks that only have one person on them. This binary countdown protocol can be used on networks with only one person. However, it's unclear how WiDom can be used to meet the communication needs of TOD in networks with multiple hops, even though this is the case. WiDom can help when there are a lot of broadcast domains [151]. Wireless networks with more than one person can't use WiDom because it doesn't deal with network-wide synchronization, media arbitration, or value transfer [152]. Multi-hop tick synchronization and a new collision-protected network-wide medium arbitration algorithm with a fixed latency are the parts of our method that make it very precise and easy to use.

4.2.6. Random access protocol

In a random-access protocol, each station can send data simultaneously on one channel at a time. Nobody can control or depend on anyone else. Nobody can control or depend on anyone else [153]. Wireless devices use a random-access channel (RACH) to connect to the mobile network (TDMA/FDMA and CDMA-based networks) to talk to each other. This channel is used to set up calls and send quick data bursts, like when users text. Revenue Adjustment Charges are set off when a mobile device moves, called MO (Mobile Originating). Four protocols let more than one person get on the same network simultaneously [154]. ALOHA, CSMA/CD, and CSMA/CA are the four types of protocols. Carrier Sense Multiple Access (CSMA): Network signals are picked up or seen before they are sent out in this access method to connect Ethernet devices. This method is used when there are a lot of computers or other devices that need to be connected, and as long as it's called the "CSMA protocol," it's part of the "MAC (Media Access Control) protocol [155].

It's part of the medium access layer in communication networks when random access protocols are used [156]. In the 1970 s, random access systems first came out. Packet collisions are now seen as a good thing, not a bad thing. This is because new solutions, primarily based on the concept of successive interference cancellation, have been developed. This is a significant change from the way things were before. A new study has enabled random-access protocols to match or beat coordinated protocols' performance [157].

4.3. How the network topology affects network communication?

It's called a "network" when two or more devices communicate through links in the network. They can talk and share resources and information [158–161]. People and things that make up a network are:

Device: There are many different types of devices. They are things that connect to a network and can send and receive data. There are computers, printers, and cell phones, but many more.

Communication link: Devices on the network can communicate through this method. When using radio waves to connect, users make a physical connection. Copper wire or optical fiber can also be used to make a connection.

- Reasons for establishing a network:
- It facilitates network and processor sharing.
- This router type offers the best performance and the lowest cost.
- The software and hardware must work together.
- A person in charge of network resources.

4.3.1. Types of networks

- A computer network can transmit data by two methods:
- Point-to-Point: There is a unique way for the two devices to talk. Only the two linked things are taking up all this space because no other devices are connected.
- Multipoint (Broadcast): Several devices can use the same link simultaneously, but it's not always possible. They all take up the same amount of space.

4.3.2. Network topology

This refers to how a network's nodes and links are put together or how it looks. Physical and logical are two types of network topologies that are types of network topologies [162]. When a network is set up, how it is laid out is significant to how it works. Bus topology, Ring topology, Star topology, Mesh topology, and Tree topology are several types of network topology.

- It is essential to think about network topology when people are building a network.
- The topology chosen will affect how well the network works.
- The network's topology determines the media used to connect the network.
- The cost of the network will be affected by how it is set up.

The network arrangement of nodes, devices, and links is known as its topology. Every topology has both good and bad things about it. So, the best topology will be determined by how big and how many people use the network [163]. Reliability, security, cost, and scalability are all things to consider when choosing a network topology.

4.4. Several issues when designing the ad hoc network routing protocol

In this section, several issues while designing the Ad Hoc network routing protocol have been described below [164–167]:

4.4.1. Mobility of nodes

Nodes in a wireless network that aren't set up in a particular way move around significantly, which changes the network's topology. This causes a lot of path breaks in the session that is going on. There is a lot of noise when intermediate nodes in the path or end nodes move. In wired networks, where all the nodes stay the same, these things don't happen because of their strong links. Wired network protocols find new routes when a path is broken, but it takes a long time to figure out how to get back. Wireless networks that move a lot, wired network routing protocols can't be used. This is because the topologies of these networks change a lot. After all, nodes move around. Ad hoc wireless network routing protocols must manage mobility efficiently and effectively, so they must be able to do this well.

4.4.2. Bandwidth constraints

Wired networks now have a lot of room for data to move because of fiber optics and wavelength division multiplexing (WDM) technologies, making it easier for data to move. Because the radio band in a wireless network is tiny, it can send and receive much less data than a wired network. Routers must use the maximum bandwidth while decreasing how long they take. Because routing protocols have limited bandwidth, they can't keep track of where things are. There is a lot of control overhead because the topology changes so often. They aren't suitable for ad hoc wireless network routing because they need to know the whole topology of the network.

4.4.3. Error-prone shared broadcast radio channel

People in ad hoc wireless networks must deal with a unique problem because the radio channel is broadcasting. Wi-Fi links have time-dependent characteristics like link capacity and link error rate. This means that the ad hoc wireless network routing protocol and the MAC layer must work together to find better routes. In ad hoc wireless networks, there are also problems with data and control packets getting mixed up. This is because of a problem with the terminals that are hidden. To set up an ad hoc wireless network, routing protocols look for less crowded routes.

4.4.4. Hidden and exposed terminal problems

It is referred to as the "hidden terminal problem" when two nodes that aren't in direct range but are in the range of another send simultaneously. When two nodes send packets simultaneously, the people involved are strangers. They are not acquainted. There are numerous methods for avoiding this issue, including medium access collision avoidance for wireless (MACAW), long-term acquisition multiple access, and dual busy tone multiple

access (DBTMA). It's important to note that while this may not completely solve the issue, the likelihood of collisions is decreased [168].

4.4.5. Resource constraints

They're having trouble getting enough battery life and processing power, which are critical but limited resources. The devices used in ad hoc wireless networks must typically be small, light, and have a power source. As for battery and processing power increase, nodes become more significant and less portable, making them more challenging to move. If users don't have a lot of resources, they should make ad hoc wireless network routing protocols as efficient as possible.

5. Wireless mesh network

A new technology called mesh networking is the most recent in wireless internet technology, and it's called that, too. If users set up a mesh network, users should get Wi-Fi all over their house. A Wi-Fi mesh network has a lot of mesh nodes (also called points, mesh extenders, or satellites) that connect. These mesh nodes work together to make an extensive Wi-Fi network for one home [169–171]. Many nodes in Wi-Fi mesh networks talk to each other to use the same wireless connection over a more extensive area. The main mesh Wi-Fi router is one of the nodes. Putting base stations around a home or small office to ensure there aren't dead and slow spots where phones don't work would also be done away. Those promises were kept a few years after it was first made public [172]. Mesh networks are the best choice for people who want to set up a new network more extensive than one Wi-Fi gateway can handle because they don't need a single one. Routers and switches can get the internet when connected to the modem. They can then send Wi-Fi to their devices to use the internet [173]. As many start-ups started making mesh nodes, people began making network devices with wireless radios and software to form an overlapping network without coordinating a single person. Access points or gateways are not crucial in mesh networking.

In most cases, if a node has two or three radio systems, it has software that lets it talk to other nodes [174]. A picture of the whole network can be made even if some nodes aren't in the same place. A Wireless distribution system was used to connect base stations wirelessly in the early days of Wi-Fi. Two people can get different kinds of information when they use a WMN. Each node has at least one, often many, paths to other nodes [175–177]. The network is more stable when the WAP or connection breaks down because it doesn't depend on the WAP or connection. There could be a topology for a wireless mesh network that is full or part mesh. It is only possible for nodes in a partial mesh architecture to talk to other nodes in their immediate area. Two nodes don't talk to each other when data is sent to another [178,179]. It has to move from one node to another to get the customer to transmit data. Users can keep the network going by getting data from one node and sending it to the next node, called a mesh cloud. Networks with more nodes are more extensive, meaning more mesh client devices can connect to the internet.

6. Wireless sensor network

A wireless sensor network, or WSN, is a group of sensors placed in different places worldwide and sending information to a central point. Things that WSNs can measure include things like the temperature, the sound level, the pollution level, the humidity level, the wind, and the temperature of the air [180–182]. Many of the things these networks have in common with ad-hoc networks are also valid. In both industrial and consumer applications, these

networks are used, for example, to keep an eye on industrial processes and keep an eye on the health of machines. A WSN comprises nodes that range from a few to hundreds or thousands, and they all connect. These nodes have the same parts: a microprocessor, a battery, energy source [183–185]. One of the most common parts is an antenna built into a radio transceiver. It makes the range and coverage area of the network bigger by setting up an ad-hoc network.

WSNs can be simple, like a single-star network, or very complicated, like a many-hop wireless mesh network. Routing or flooding can get things to move around a lot more. In a WSN, sensors spread over an extensive area work together to send their data to one place called a backbone [186–188]. They watch things like temperature, sound, vibration, pressure, humidity, mobility, or pollution and send their data together. Modern networks are two-way, which means that other people can control sensors. The internet and the rest of the world will only need one node to join them. People now use wireless networks every day, and a lot use them. Ad hoc networks are better than wired networks in many ways. Some don't need much infrastructure, are easy to put together, and can be changed quickly [189–191]. Fig. 22 shows a lot of different requirements of wireless sensor networks.

7. Discussion and comparison table

This section compares mobile ad hoc networks, wireless mesh networks, and wireless sensor networks over different aspects such as application, characteristics, limitations, challenges, routing protocols, pros, and cons.

Table 3 describes the comparison between MANET, WMN, and WSN based on application. The conclusion arises that MANET is more applicable because it can be used in all sectors than WSN and WMN.

Table 4 describes a comparison between MANET, WMN, and WSN based on characteristics, and the conclusion arises that MANET has fewer limitations or was bandwidth limited. Hence, it is more applicable to make as per user required. In contrast,

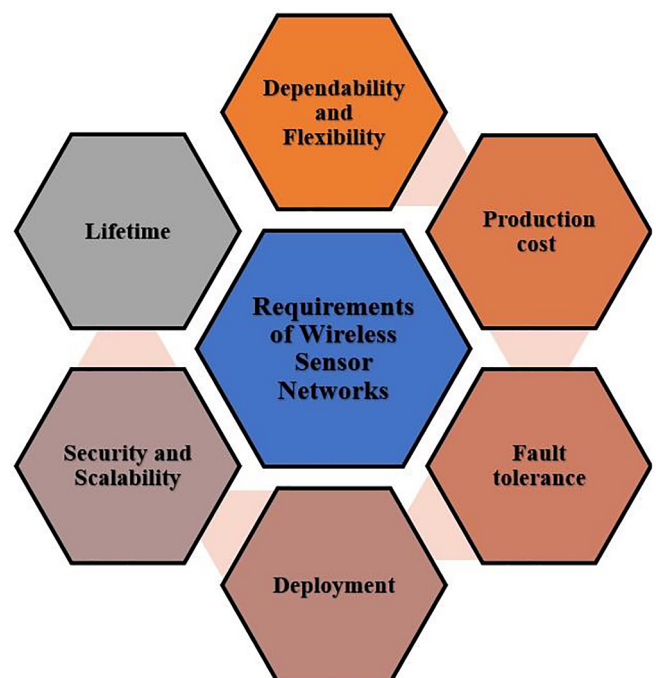


Fig. 22. WSN Requirements.

Table 3
Application of MANET, WMN, and WSN [192–198].

MANET	WMN	WSN
Gaming Service Military Service Commercial Use Industrial Sector Education Sector Entertainment Sector	Communication Need Battlefield, Surveillance High-Speed Mobile Voice over Internet Protocol Smart Home Devices Electric Smart Meter	Internet of Things Agriculture and Landslide Detection Noise Level of the Surrounding Surveillance and Monitoring Medical Application, Humidity Environmental temperature

Table 4
Characteristics of MANET, WMN, and WSN [199–205].

MANET	WMN	WSN
Variable Capacity Links Bandwidth Constrained, Autonomous Behaviour Energy Constrained Operation, Limited Security Less Human intervention, Limited Bandwidth No Centralized Control, Dynamic Topologies, Random Change in Devices	High Reliability Random Deployment Strong Mobility Non-line-of-sight (NLOS) Transmission High Data Bandwidth and Fast Movement Security and Confidentiality	Application-Specific Design Resource Constraints and Dynamic Topologies Node Failure and Unreliable communication Communication Models Scalability and Density Communication Paradigm

Table 5
Limitations in MANET, WMN and WSN [206–212].

MANET	WMN	WSN
Bandwidth Constraints, Transmission error Processing Capability, Security	Global Positioning System, General Packet Radio Service Electrocardiogram, Monitoring	Hardware and Operating System, Deployment, Localization
Energy Constraints, Scalability Problems High Latency, Location, Roaming Commercially Unavailable, Limited Resources	Telemonitoring of Surgical Procedure Mobile Emergency Management Preoperative Information Management	Wireless Radio Communication Characteristics, Network Layer, Transport layer Medium Access Scheme, Calibration, Synchronization Middleware, Data Aggregation, and Data Dissemination Quality of Service, Programming Models for Sensor Network, Security
No Central Check on the Network	Emergency Telemedicine	Architecture, Database Centric, and Querying

Table 6
Routing protocols in MANET, WMN, and WSN [213–219].

MANET	WMN	WSN
Proactive Routing Protocol Reactive Routing Protocol Geographical Routing Protocol Hybrid Routing Protocol Hierarchical Routing Protocol	Associativity Based Routing Protocol, Predictive Routing Based Protocol Hazy Sighted Link State Protocol, Zone Routing Protocol Hybrid Wireless Mesh Protocol, Temporally Ordered Routing Algorithm Protocol Infrastructure Wireless Mesh Protocol, Dynamic Source Routing Protocol Order One Routing Protocol, Ad-Hoc On-Demand Distance Vector	Featured Routing Protocol Energy-Efficient Routing Protocol Delay Less Routing Protocol Secure Routing Protocols Reliable Routing Protocols

MANET and WMN is not application-specified design, and MANET has variable capacity characteristics while WMN and WSN are not. WMN has attributes of security, confidentiality, and muscular mobility.

Table 5 describe the comparison between MANET, WMN, and WSN based on limitation. The conclusion arises that MANETs have many restrictions like bandwidth constraints, security, etc., whereas WMN does not have security limitations. WSN has limitations in the network layer, transport layer, and many more, while WMN has limitations in GPS, GPRS, and the health sector, a significant restriction in WMN. In contrast, WSN has limited architecture design, Data aggregation, and localization, whereas MANET has energy constraints and scalability problems.

Table 6 describes the comparison between MANET, WMN, and WSN on routing protocols. The conclusion is that MANET, WMN, and WSN have different routing protocols where the AODV protocol works with MANET and WSN.

Table 7 describes the comparison between MANET, WMN, and WSN based on pros, and the conclusion arises that WSN has an advantage in cost, maintenance, flexibility, scalability, and security. While using WMN, wireless WiFi in the home has an advantage, making it bright. This also makes smart city ensure data security and supports redundant backup. In contrast, MANET has the advantage of separation from the administration in MANET each node.

Table 8 describes a comparison between MANET, WMN, and WSN based on challenges, and the conclusion arises that MANET is not challenging in security. At the same time, WMN has challenges in security, scalability, and coverage, while WSN has challenges in quality of services, safety, and performance issue.

Table 9 describes the comparison between MANET, WMN, and WSN on pros and concludes that MANET has disadvantages of authorization facilities, limited resources, more prone to attacks, and limited physical security. In contrast, WMN has little process-

Table 7
Pros of MANET, WMN, and WSN [220–225].

MANET	WMN	WSN
Separation from the administration of the central network	Supports redundant backups to ensure data security in the event of disc failure.	Installation, Cost, and Maintenance
Each node can perform both functions, i.e., the router demonstrating its self-sufficient nature.	Each node is connected to several others, so the neighbors find another route if one fails.	Flexibility and scalability
Nodes that configure themselves and heal themselves do not require human intervention.	Effectively cover home Wi-Fi mesh networks without compromising bandwidth.	Connectivity and Security

Table 8
Challenges in MANET, WMN, and WSN [226–232].

MANET	WMN	WSN
Lack of Centralized Entity, Reliability, Survivability, and Availability	High Performance	Quality of Service
All communication Carried over Wireless Medium, i.e., limited wireless transmission range and broadcast nature of the wireless medium	Scalability	Security and Performance
The hidden terminal problem, Exposed Terminal Problem, Ease of snooping on wireless transmission	Security	Issue
Packet losses due to transmission errors, Manageability of the network	Physical Attack, Denial of Service Attack, Black Hole Attack, and Warm Hole Attack	Energy Efficiency
Mobility-induced route changes	Coverage	Network Throughput
Mobility-induced packet losses	Interference due to External Networks	Cross-Layer Optimization
Battery Constraints, Potentially Frequent Network Partitions	Power Transmission Level, Provisioning	Scalability to the large scale of deployment
		Ability to cope with node failure

Table 9
Cons of MANET, WMN and WSN [233–239].

MANET	WMN	WSN
Resources are limited because of things like noise and interference.	In networks with limited processing power, data must frequently hop between nodes.	It is easy for hackers to get into it because users can control how waves move through the air.
Lack of authorization facilities.	Mesh networks lack a central server, making them more challenging to monitor.	Comparatively slow communication and a lot of money at the same time
More prone to attacks due to limited physical security.	Setup is complex, and node devices are more expensive than router/modem Wi-Fi networks.	WSN is distracted by objects like Bluetooth.

ing power, making monitoring more challenging; setup is complex. In comparison, WSN has the burden that it can be easily hacked, slow communication, and most importantly, distracted by objects like Bluetooth, and need a lot of money simultaneously.

8. Conclusion

Researchers discussed key management and wireless ad-hoc network routing. As evidenced by the problems these networks face and the solutions on the way, ad-hoc networking is still a relatively new field of study. Key management strategies are still too expensive and ineffective. To meet the high standards of today's networks, they need to be made more secure and durable. They can be set up in various ways, are simple to use, and quickly set up. As a result, a more significant number of people are likely to use them.

8.1. Conclusion

This paper is a brief overview of ad hoc networks, including the types of mobile ad hoc networks, wireless sensor networks, and wireless mesh networks, as well as the motivation for studying ad hoc networks, the scope of this work, the vulnerabilities of ad hoc networks, work done in ad hoc networks from 2000 to 2022, and various features and advantages of these networks. Other MANET networks are also covered in depth to clarify relocating network nodes. As a result, studying this network will aid in under-

standing ad hoc networks and comparing mobile ad hoc networks, wireless mesh networks, and wireless sensor networks on various levels such as restrictions, challenges, characteristics, advantages, disadvantages, etc. The future focus of this research study will be on improving the accuracy and effectiveness of communication between these various networks.

8.2. Future direction

There are two issues to worry about:

- To make it possible for people to talk to each other,
- To improve and add to wireless access networks.

A lot of wireless devices are likely to be around soon. Ad hoc network functions like self-configuration and independence from current networks are the most important things to consider. Personal area networks, body area networks, home networks, sensor and actuator networks, and vehicle-to-vehicle networks will be used in the future. It will be possible to do many new things because of these new types of ad hoc networks. It can also help existing wireless technologies, especially for multi-hop communication, where ad hoc networking can help. In general, when nearby terminals talk to each other without the help of base stations a long way away, there is less interference, which means the system's overall capacity goes up. This could help cut the cost of wireless access infrastructure, but it could also make networking more

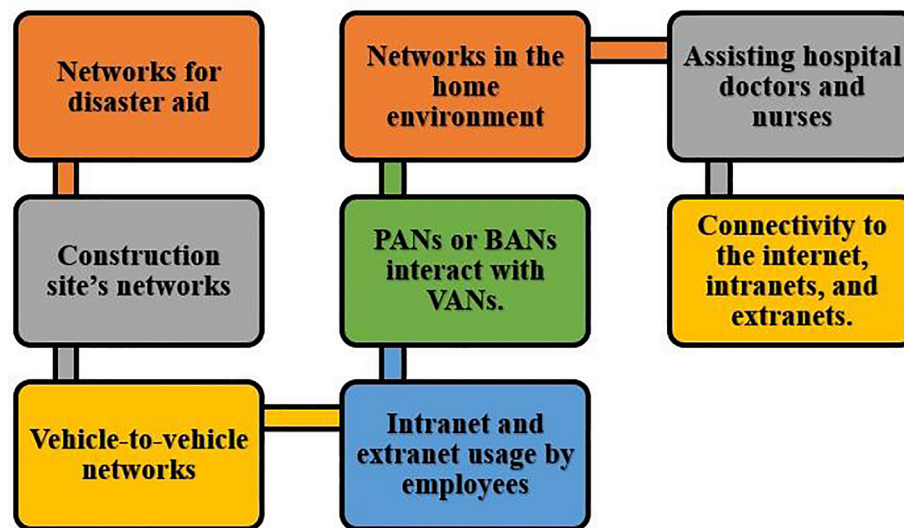


Fig. 23. Future Directions for Research of Ad Hoc Network.

complex, so it might not be the best idea. This strategy isn't just good for cell phones. It also works well for other types of networks. It's also good that Wireless Local Area Networks (WLAN) that use a lot of radio waves can be hard to see, so it's also good. People could get into the Universal Mobile Telecommunications System by setting up their networks to connect. Ad hoc networks are things like self-configuration and finding other people to connect with. To say this another way, these networks will play a big part in making wireless networks more compatible, which the future will be all about. They can be used for many different things. Some of them are shown in Fig. 23.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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